



ViFi: A Video Finding System at Video Browser Showdown 2025

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Abstract. This paper presents ViFi - a Video Finding System for Video Browser Showdown 2025. Our retrieval system is mainly based on the SigLIP, a most recent and robust visual-textual embedding model, and significantly outperforms CLIP in text-to-image retrieval on the MSCOCO data. Instead of using the given extracted keyframe from the dataset like other previous approaches, we found that the given keyframe might have missed some of the moments in the textual query. Thus, our system operates using keyframes derived from our custom keyframe extraction stage. While neighboring keyframes might share similar characteristics, such as objects' appearance, we also try to visualize the result in a relevant video segment extracted by TransNetv2. Moreover, our system supports temporal search with now-and-then textual input, allowing users to search for time-related information. Finally, we utilized recent tools to extract metadata from the keyframes, including detected objects, the count of these objects, and more. We then combined this information with user input to refine and filter the search results. Finally, we have designed a user-friendly interface offering both simple and advanced modes to enable faster examinations and assist novice users.

Keywords: Video Retrieval System · Interactive Retrieval System · Multimodal Retrieval

1 Introduction

Rapid advancements in technology have resulted in an exponential increase in the amount of information that people need to store. As a result, this creates significant pressure on search engines, which must be capable of retrieving the desired items within seconds or even less. In the video retrieval domain, the Video Browsing Showdown (VBS) [6] is an annual international competition focused on video content search, assessing the interactive video retrieval systems in live sessions at the MMM conference. Compared to the previous year, the data for this year's challenge is significantly larger in terms of video durations with the addition of V3C3 [5] and MVK 2.0 [9] datasets. In total, this year's challenge data

includes the whole V3C dataset [5] (including 28,450 videos with a duration of around 3801 h), the Marine Video Kit dataset [9], and the LapGynLHE dataset.

In this paper, we present ViFi, a Video Finding System for Video Browser Showdown 2025. The suggested video identification system, dubbed ViFi, incorporates the following key characteristics:

- First, our system text-to-image search embedding model is mainly based on SigLIP [12], a most recent state-of-the-art visual-textual alignment model, and has higher results in text-to-image retrieval compared to CLIP [1].
- Second, instead of using the given extracted keyframe provided by the dataset, our system uses keyframes generated from our customized keyframe extraction process. We also attempt to visualize the result by a list of relevant segments by grouping related keyframes.
- Third, we implement a temporal search mechanism, which allows users to perform text queries based on current and future information. In addition, we utilized recent tools to extract metadata, e.g., appeared objects and quantity of appeared objects, from keyframes and combined this information with user input to refine the search results.
- Lastly, we develop a user-friendly interface that features both basic and advanced modes, enabling faster human selection and aiding novice users.

2 Related Work

In recent years, numerous systems have taken part in the Video Browsing Showdown (VBS) competition. Integrating the semantics of images into the retrieval process can enhance the accuracy and relevance of search results. CLIP stands out as a powerful joint embedding model for both textual and visual data, frequently utilized for text-to-image retrieval tasks. This model has shown significant efficiency and is widely recognized by many teams in the recent VBS challenge as the visual-language embedding model. Our retrieval framework primarily relies on SigLIP [12], an enhanced version of CLIP [1]. SigLIP [12] notably surpasses CLIP in text-to-image retrieval performance on the MSCOCO dataset [3] and zero-shot classification on the ImageNet-1K dataset [2].

Concept-based search aims to improve image retrieval by tagging images with visual identifiers or concepts, such as detected objects, scene text, main colors, etc. This information is then indexed along with other metadata to match the images with the given keywords, and this search mechanism is integrated into many previous teams [4, 7]. In addition, temporal information frequently emerges in video queries. Recent systems show that applying temporal search can improve efficiency and accuracy with temporal queries [10].

3 ViFi - Our Proposed Video Finding System

3.1 System Overview

The overview of our system architecture is shown in Fig. 1. Our system’s primary operating mechanism is based on text-to-image retrieval using the

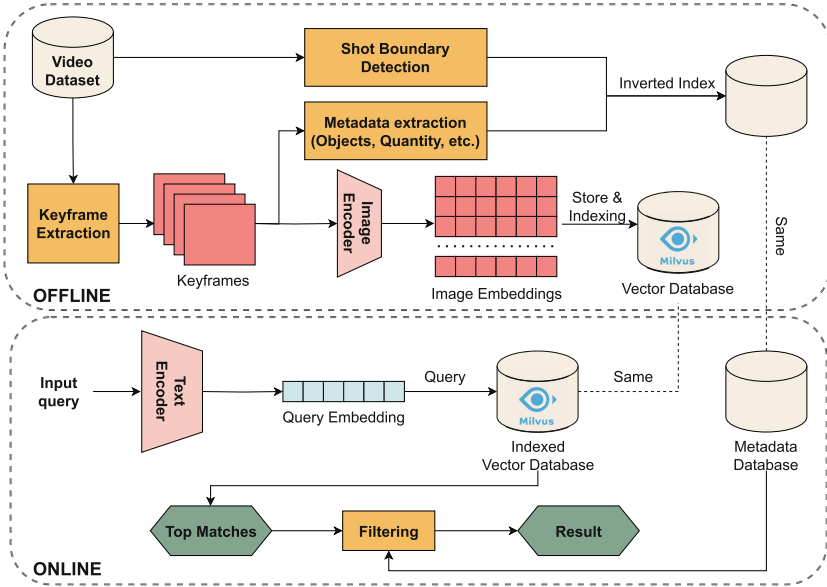


Fig. 1. The overview architecture of ViFi - our proposed Video Finding System.

visual-language embedding model and Milvus vector database. In the offline processing stage, we extract our own keyframes and the embeddings corresponding to the extracted keyframes from the given video dataset. We employ Milvus¹ as the vector database to store and index the keyframe embeddings. We also extract information in the keyframes, such as the appeared objects and quantity of appeared objects, using different tools and storing them in a different database.

3.2 Data Hierarchy

Figure 2 illustrates our system’s data hierarchy organization. Our system is based on the extracted keyframes through our customized keyframe extraction process rather than relying on the keyframes provided by the dataset. Our analysis revealed that the provided keyframe in the datasets may not capture all the moments mentioned in the textual query. For example, with the query “A herd of donkeys/mules are walking down a walkway upstairs, followed by a herdsman.” the provided keyframe in the dataset does not have any clearly related visual information about “A herd of donkeys/mules” or “a herdsman” (shown in Fig. 2 - Provided Keyframes).

While keyframes within the same segments frequently exhibit similar visual characteristics, such as object appearances, colors, lighting, and more, displaying

¹ <https://milvus.io/>

each keyframe in the result list may become redundant for information dissemination. Therefore, we also try to group the neighboring keyframes into segments and display them in the result list. Specifically, we employ TransNetv2 [8] to split the video into smaller segments and group all retrieved keyframes into related segments in the resulting interface. In the same segment, we choose the keyframe with the highest similarity score with the input query as a thumbnail.

3.3 Embedding Models

We use SigLIP [12] as the textual and visual embedding model for text-to-image and image-to-image retrieval. SigLIP is the recent state-of-the-art visual language model and achieved significantly higher results than CLIP in text-to-image retrieval on the MSCOCO data [3].



Fig. 2. Overview of our system data hierarchy. From the frames in the given videos, we extract the keyframes using our own process. Then, instead of displaying the relevant keyframes on the results screen, we use segments as representatives.

3.4 Keyframe Extraction

While the provided keyframe might have missed the interesting moments in the query, we also designed our own keyframe extraction process. In particular, we first use ffmpeg² to extract the keyframes in the given video and extract these keyframe features using SigLIP [12]. Then, we employ the sliding window technique by comparing the similarity score of two neighboring keyframe embeddings. If the similarity score is higher than a certain threshold, we eliminate one of the keyframes in the window.

² <https://www.ffmpeg.org/>.

3.5 Metadata Information Extraction

For each extracted keyframe, we use YOLOv10 [11] trained on the MSCOCO dataset [3] with 80 common categories as the object detector to detect objects in each extracted keyframe. The detected objects, as well as the quantity of each detected object, are stored in the dedicated database using an inverted index. The stored information is used to filter top-match results.

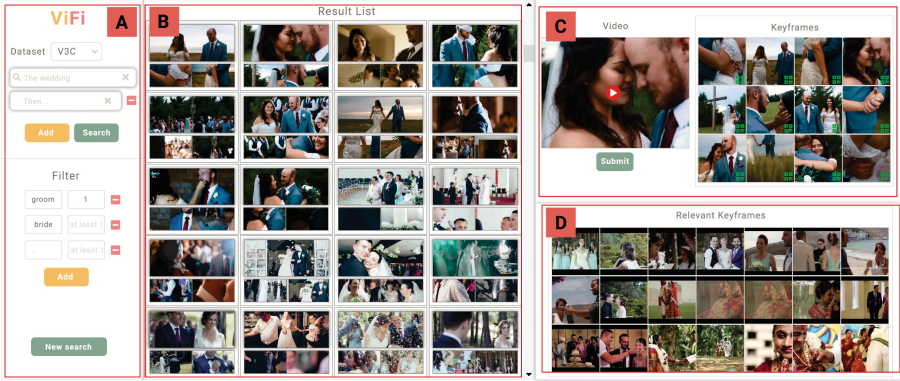


Fig. 3. The prototype user interface of ViFi.

3.6 Temporal Search

Temporal information is a crucial aspect when searching for videos and frequently emerges in user queries. Therefore, we implemented an algorithm that allows users to perform temporal searches based on now-and-then information. Specifically, our system will allow users to enter two textual inputs, one for current information and one for future information. First, the system will perform a query based on current information and return top-matched results with corresponding present similarity scores. Then, for each top match result, the system will compute a list of future similarity scores between the next k -frames through sampling of the returned frame and the future information textual query. Finally, the future similarity score is selected as the topmost score within the list of future similarity scores, and we re-rank the result based on the weighted average of the present similarity score and the future similarity score.

4 Designed User Interface

The main interface of ViFi is illustrated in Fig. 3 with four main sections: (A) Search Section, (B) Result Display Section, (C) Result Selection Interface, and (D) Similar Result Interface.

4.1 (A) Search Section

The simple version of the search section includes a textual input and a search button, which users can perform retrieval based on textual query. For the temporal search, we provide an ‘Add’ button that allows users to add additional search bars for the users to describe the next moment in the video.

The search section also includes the filter component, in which users can describe in detail the object appearing in the frame along with the quantity of the object. If the user does not specify the amount of the object, it will search for all keyframes in which that object appears.

4.2 (B) Result Display Section

When no search query has been entered in the main content area, the system displays the message ‘Empty result.’ When the user performs a new query in section (A), the retrieval results are shown in descending order based on the similarity scores within this section. The results are arranged from left to right and top to bottom, based on the relevance score from highest to lowest. Users can browse the results and find the videos that best match their search criteria.

The list of results can be a list of retrieved keyframes or relevant segments. If the result is relevant segments, the thumbnail is the keyframe with the highest similarity score. We additionally show the results retrieved with the next and previous moments.

4.3 (C) Result Selection Interface

If users wish to view the content of a segment in detail, they can select it from the results list. At this point, the interface will display information about the video containing that segment on the left. The list of keyframes in the selected video will be displayed on the right, allowing users to choose their desired keyframe easily. We also provide a "Submit" button below the video, enabling users to select and submit their answers. The submitted answer can be a specific time of the video or the keyframe chosen from the list.

4.4 (D) Similar Result Interface

This section will be displayed when the user clicks the ‘Relevant’ button on the images in the keyframes component of section C. The interface will display all keyframes that are similar to the selected keyframes. The user can also choose similar keyframes to display using the (C Result Selection Interface) or submit it directly.

5 Conclusion

In this paper, we provide an overview of our development of ViFi - a Video Finding System for the 2025 Video Browser Showdown. Our system is mainly based

on SigLIP, a visual-language alignment approach that recently outperformed CLIP, as our embedding model. Our system functions by utilizing keyframes obtained through our specialized keyframe extraction process. Moreover, our system supports temporal search and filter mechanisms based on objects and the number of objects appearing in the video. Finally, we design an intuitive interface that allows quicker human selection and supports novice users.

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References

1. CLIP: Connecting text and images. <https://openai.com/research/clip>
2. Deng, J., Dong, W., Socher, R., Li, L.J., Li, K., Fei-Fei, L.: Imagenet: A large-scale hierarchical image database. In: 2009 IEEE Conference on Computer Vision and Pattern Recognition, pp. 248–255 (2009). <https://doi.org/10.1109/CVPR.2009.5206848>
3. Lin, T.Y., et al.: Microsoft coco: common objects in context. In: Fleet, D., Pajdla, T., Schiele, B., Tuytelaars, T. (eds.) Computer Vision - ECCV 2014, pp. 740–755. Springer International Publishing, Cham (2014)
4. Pantelidis, N., et al.: VERGE in VBS 2024, pp. 356–363 (01 2024). https://doi.org/10.1007/978-3-031-53302-0_32
5. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3c—a research video collection. In: MultiMedia Modeling: 25th International Conference, MMM 2019, Thessaloniki, Greece, January 8–11, 2019, Proceedings, Part I 25. pp. 349–360. Springer (2019)
6. Schoeffmann, K., Lokoč, J., Bailer, W.: 10 years of video browser showdown. In: Proceedings of the 2nd ACM International Conference on Multimedia in Asia. MMAsia '20, Association for Computing Machinery, New York, NY, USA (2021). <https://doi.org/10.1145/3444685.3450215>
7. Schoeffmann, K., Nasirihaghighi, S.: DiveXplore at the Video Browser Showdown 2024, pp. 372–379 (01 2024). https://doi.org/10.1007/978-3-031-53302-0_34
8. Souček, T., Lokoč, J.: Transnet v2: An effective deep network architecture for fast shot transition detection. arXiv preprint [arXiv:2008.04838](https://arxiv.org/abs/2008.04838) (2020)
9. Truong, Q.T., et al.: Marine video kit: A new marine video dataset for content-based analysis and retrieval. In: MultiMedia Modeling - 29th International Conference, MMM 2023, Bergen, Norway, January 9–12, 2023. Lecture Notes in Computer Science, Springer (2023)
10. Vadicamo, L., et al.: Evaluating performance and trends in interactive video retrieval: Insights from the 12th vbs competition. IEEE Access **12**, 79342–79366 (2024). <https://doi.org/10.1109/ACCESS.2024.3405638>
11. Wang, A., et al.: YoloV10: Real-time end-to-end object detection. arXiv preprint [arXiv:2405.14458](https://arxiv.org/abs/2405.14458) (2024)
12. Zhai, X., Mustafa, B., Kolesnikov, A., Beyer, L.: Sigmoid loss for language image pre-training. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 11975–11986 (2023)